

Civic Services Surveillance & Monitoring

On-device YOLO11n detection of 7 civic issues — dataset sources and training pipeline

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Civic Services Surveillance & Monitoring — real-time, on-device detection of 7 civic issues:

- 1 Garbage on road (>1m accumulations)
- 2 Road potholes (>10 inch)
- 3 Open / broken manholes
- 4 Non-working streetlights
- 5 Encroachments (unauthorized occupation/obstruction)
- 6 Billboard & hoarding compliance
- 7 Cattle on road

Each detection auto-generates an incident report: image, camera details, timestamp.

Approach & Architecture

- **Single unified detector**, not 7 separate models — one YOLO11n running all 7 classes simultaneously is far cheaper on a phone than 7 models.
- **Model**: YOLO11n (Ultralytics) — chosen over YOLOv8n/YOLOv10n/YOLO-NAS/MobileNet-SSD/EfficientDet-Lite/NanoDet for the best size/speed/mobile-export tradeoff. ~5–6MB, one-command export to TFLite/ONNX/CoreML.
- **Deployment**: native Android app (CameraX + TFLite), fully offline — no internet required for detection.
- **Streetlight state**: no dataset anywhere labels lit/unlit streetlights, so the model only localizes the pole; a brightness heuristic classifies lit/unlit at night.
- **Reporting**: every confirmed detection saves a cropped image + JSON (class, confidence, timestamp, camera ID) automatically.

Dataset Pipeline Overview

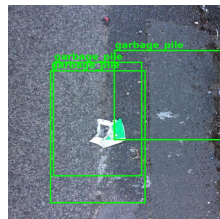
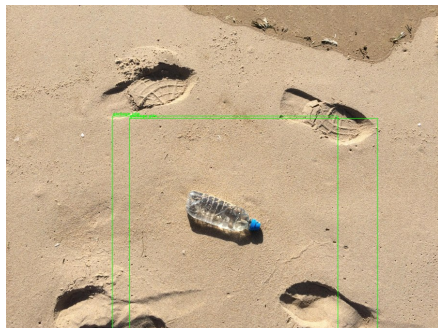
- Datasets sourced from **Roboflow Universe**, filtered/remapped per class into one unified YOLO-format dataset.
- **16,766 images** total, split **80.1% train / 19.9% val** (13,435 / 3,331 images).
- Split ratio inherited from each source's own train/valid partition — not a custom stratified split.
- No independently held-out test set beyond the val split used for monitoring.

Class	Images	% of dataset
garbage_pile	6,350	37.9%
pothole	3,753	22.4%
billboard_hoarding	3,399	20.3%
streetlight_pole	1,232	7.3%
manhole_open_broken	964	5.8%
encroachment	621	3.7%
cattle	447	2.7%

garbage_pile — 6,350 images

Sources:

- **TACO** — 3,572 img
<https://universe.roboflow.com/mohamed-traore-2ekkp/taco-trash-annotations-in-context>
Non-India (international litter photos)
- **Trash Detection (Roboflow)** — 2,778 img
<https://universe.roboflow.com/yolo-hfplh/trash-detection-1fjjc-nd1dg>
Origin unconfirmed



pothole — 3,753 images

Source:

- **Pothole Detection — Intel Unnati Training Program** — 3,753 img
<https://universe.roboflow.com/intel-unnati-training-program/pothole-detection-bqu6s>
India (Intel Unnati is an India-based program)

Best-performing detection-only class besides cattle — AP50 0.85.



manhole_open_broken — 964 images

Sources (filtered to broken/open/uncovered classes only):

- **Manhole Cover Dataset YOLO** — 467 img
<https://universe.roboflow.com/create-dataset-for-yolo/manhole-cover-dataset-yolo>
- **Open-Manholes** — 444 img
<https://universe.roboflow.com/aibased-solution-for-realtime-detection-of-road-and-open-manholes>
- **Manhole Broken v2** — 53 img
<https://universe.roboflow.com/2nd-workspace/manhole-broken>
- All origins unconfirmed



streetlight_pole — 1,232 images

Source:

- **Street Light (Sashank S)** — 1,232 img
<https://universe.roboflow.com/sashank-s/street-light>
Origin unconfirmed

Note: localizes the pole only — no dataset anywhere labels lit/unlit state. On/off is inferred at inference time via a brightness heuristic on the cropped region at night, not a learned label.



encroachment — 621 images

Sources:

- **Street Vendors (public)** — 495 img
<https://universe.roboflow.com/public-2mrlb/street-vendors>
- **Street Vendors India** — 59 img
<https://universe.roboflow.com/mit-f83x1/street-vendors-india>
India
- **Footpath Guide** (filtered to Vendor/Vendor cart/Obstacle/Street goods) — 65 img
<https://universe.roboflow.com/vendor/footpath-guide>
- **Illegal Parking** (original) — 2 img

Weakest class: started with only 2 usable images (AP50 0.00); after adding these sources, AP50 improved to 0.35.

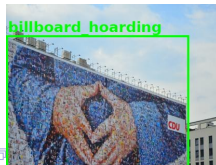
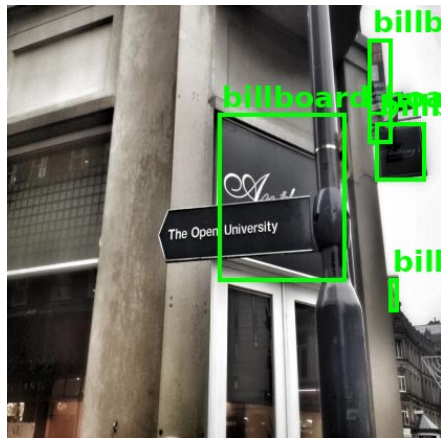


billboard_hoarding — 3,399 images

Source:

- **Billboard Object Detection (Roboflow)** — 3,399 img
<https://universe.roboflow.com/arслан-ongr8/billboard-xl-vz1>
Origin unconfirmed

Note: detects billboard/hoarding presence only — no dataset provides ground-truth permit or size-compliance labels, so “compliance” checking is not implemented, only localization.

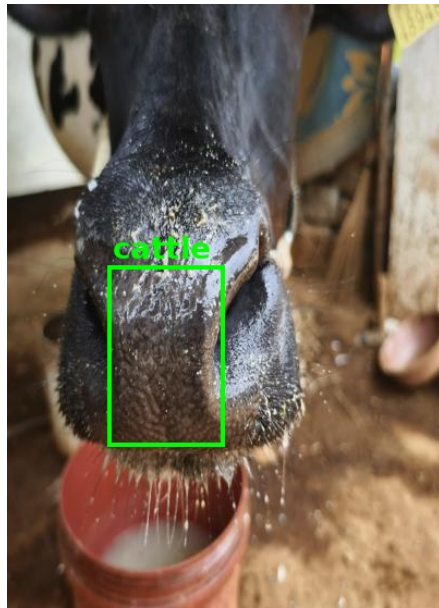


cattle — 447 images

Source:

- **Cow Indian (Roboflow)** — 447 img
<https://universe.roboflow.com/sakshi-kumari-70uxi/cow-indian>
India (explicitly “stray cattle on Indian roads”)

Best-performing class overall — AP50 0.99. COCO/Open-Images pretrained weights already include a cow class, giving this class a head start.



Training Results — Run 1 vs. Run 2 (after adding encroachment data)

Class	Run 1 AP50 (16,094 img)	Run 2 AP50 (16,766 img)
cattle	0.992	0.986
pothole	0.849	0.851
manhole_open_broken	0.823	0.843
billboard_hoarding	0.689	0.684
streetlight_pole	0.650	0.661
garbage_pile	0.608	0.643
encroachment	0.000	0.350
Overall mAP50	0.659	0.717
Overall mAP50-95	0.420	0.472

Both runs: YOLO11n, 100 epochs. Adding encroachment data fixed that class entirely without hurting any other class.

Dataset Origin — India vs. Non-India

	Images (% of 16,766)
Confirmed India	4,259 (25.4%)
Confirmed Non-India	3,572 (21.3%)
Unconfirmed origin	8,935 (53.3%)

- Only 3 of 15 source datasets are confirmed Indian: Intel Unnati (pothole), Street Vendors India (encroachment), Cow Indian (cattle).
- Most Roboflow community uploads never state geographic origin.
- Classes with confirmed Indian data (pothole, cattle) are also the best-performing — plausible but unproven relationship.
- Real-world accuracy on Indian roads for garbage/manhole/streetlight/billboard may be lower than validation numbers suggest.

Android App

- Native app: CameraX + TFLite, fully on-device, works with **no internet**.
- Live detection overlay with FPS/detection counter.
- Manual photo capture and video recording alongside live detection.
- Automatic incident reporting: cropped image + JSON per confirmed detection, 60s per-class cooldown to avoid duplicate spam.
- Landing page, Reports gallery, About screen with light/dark mode toggle.
- Verified live on a Samsung Galaxy S24 via adb — camera rotation and model input-layout bugs were found and fixed by testing against real recorded video, not by code inspection alone.

Known Limitations

- No physical size/dimension measurement — garbage ($>1\text{m}$) and pothole ($>10\text{in}$) thresholds from the problem statement are not verified, only generic object detection.
- No alerting/notification pipeline — reports save locally to the phone only.
- No billboard compliance logic (permits, size rules) — detection only.
- Streetlight lit/unlit state is a brightness heuristic, not a trained classifier.
- Encroachment (AP50 0.35) is still the weakest class — only 621 training images.
- No independently held-out test set beyond the validation split.

Thank You

Questions?